RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 14/01/2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : V	**UG**	**Duration : 30 + 90 Min**
Course Title: Enterprise Information Systems		**Course Code : IM355TBB**

Note:
1. Answer all the questions

Sl. No	Questions	M	BT	CO
	Part – A			
1.	List the three main streams that constitute a Supply Chain Management flow.	1	1	CO1
2.	State how an EIS system provides a basic operational prerequisite for global supply chain operations.	1	2	CO1
3.	Give two examples of common cybersecurity threats specific to an EIS environment.	2	2	CO4
4.	Define Agile Project Management based on its delivery characteristics.	2	2	CO4
5.	List two primary operational advantages a business gains by adopting a Cloud-based EIS (SaaS model).	2	2	CO4
	Part – B			
1	Define the terms “Web-Enabled ERP” and the “Extended Enterprise.” Explain why the transition to web technology was essential for maximizing the value of ERP in a supply chain context.	10	2	CO1
2	Clearly state the primary scope and focus of a standalone Enterprise Resource Planning (ERP) system versus a dedicated Supply Chain Management (SCM) system.	10	2	CO2
3	Define Artificial Intelligence (AI) from an Enterprise Information Systems (EIS) perspective. List and briefly explain four distinct business functions within an EIS (e.g., SCM, CRM) where AI is being successfully deployed.	10	1	CO4
4	What is the Internet of Things (IoT)? And explain its foundational role in modernizing Enterprise Resource Planning (ERP) systems. Detail how IoT enhances real-time visibility in production and supply chain monitoring.	10	1	CO4
5	Discuss the concept of “Cloud-Native” EIS architecture. Explain why this approach, utilizing micro-services and APIs, is considered more flexible and scalable than traditional monolithic cloud migration methods.	10	2	CO4

Academic year 2024-2025(Odd Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	**29th January 2025**	**Maximum Marks**	**10+50**
Course Code	**IM255TBB**	**Duration**	**120 Min**
Sem	**V Semester**	**Improvement CIE**	
ENTERPRISE INFORMATION SYSTEMS			

Note:

1. Answer all the Questions.

PART – A

Sl. No	**Questions**	**M**	**BT**	**CO**
1	Define Business Process Automation (BPA).	02	L2	CO3
2	Name two common tools used in Robotic Process Automation (RPA).	02	L2	CO3
3	Mention one advantage of implementing automation in business processes.	02	L1	CO2
4	What is a key characteristic of structured data?	02	L1	CO1
5	What are the three types of data analytics?	02	L2	CO2

PART – B

Sl. No	**Questions**	**M**	**BT**	**CO**
1.	A mid-sized manufacturing company faces delays in order processing due to repetitive manual tasks such as data entry and validation. Propose a solution using Robotic Process Automation (RPA). Discuss how implementing RPA can streamline the process and improve efficiency.	10	L3	CO2
2.	A retail company plans to automate its invoice processing system using RPA. Explain the steps involved in the RPA implementation process for this scenario.	10	L3	CO3
3.	Design a simplified approach to integrating Business Process Automation (BPA) in the inventory management process of an e-commerce company. Include key steps and expected outcomes.	10	L4	CO4
4.	A retail chain plans to improve its inventory management system using data warehousing. Explain how a data warehouse can help in managing inventory more effectively. Include key components of a data warehouse and their roles.	10	L3	CO3
5.	Explain how data analytics in Enterprise Information Systems (EIS) can be used to improve the decision-making process for a manufacturing company aiming to reduce operational costs.	10	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz Marks	02	04	04	–	04	06	–	–	–	–
	Test Marks	–	10	20	20	–	10	30	10	–	–

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**29th January 2025**	**Maximum Marks**	**50**
Course Code	**IM355TBD**	**Duration**	**90 Min**
Sem	**V**	**CIE – Improvement**	
ADVANCED DECISION MODELLING (Professional Core Elective)			

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part A			
1.	List two functional roles of inventory.	02	1	1
2.	Identify two key factors in inventory problem analysis.	02	2	3
3.	How does an EOQ model without shortages work?	02	2	2
4.	How does inventory contribute to production efficiency?	02	2	2
5.	How is Markov Analysis used in business decision-making?	02	2	3
	Part B			
1.	The production department of a company requires 3,600 kg of raw material for manufacturing a particular item per year. It has been estimated that the cost of placing an order is Rs 36 and the cost of carrying inventory is 25 per cent of the investment in the inventories. The price is Rs 10 per kg. Help the purchase manager to determine an ordering policy for raw material.	10	1	1
2.	Each unit of an item costs a company Rs 40. Annual holding costs are 18 per cent of unit cost of the item due to miscellaneous changes: 1 per cent for insurance, 2 per cent allowances for obsolescence, Rs 2 for building overheads, Rs 1.50 for damage and loss, and Rs 4 miscellaneous costs. The annual demand for the item is constant at 1,000 units. Placing each order costs, the company Rs 100. i. Calculate EOQ and the total costs associated with stocking the item. ii. If the supplier of the item will only deliver batches of 250 units, how are the stock holding costs affected? iii. If the supplier relaxes his order size requirement, but the company has limited warehouse space and can stock a maximum of 100 units at any time, what would be the optimal ordering policy and associated costs?	10	2	2
3.	A product is sold at the rate of 50 pieces per day and is manufactured at a rate of 250 pieces per day. The set-up cost of the machines is Rs 2,000 and the storage cost is found to be Re 0.15 per piece per day. With labour charges of Rs 3.20 per piece, material cost at Rs 2.10 per piece and overhead cost of Rs 4.10 per piece, find the minimum cost batch size if the interest charges are 8 per cent (assume 300 working days in a year). Compute the optimal number of cycles required in a year for manufacturing of this product.	10	3	3

4.	A commodity is to be supplied at a constant rate of 200 units per day. Supplies of any amount can be obtained at any required time, but each ordering costs Rs 50; cost of holding the commodity in inventory is Rs 2 per unit per day while the delay in the supply of the item induces a penalty of Rs 10 per unit per day. Find the optimal policy (Q, t), where t is the reorder cycle period and Q is the inventory after reorder. What would be the best policy to adopt if the penalty cost becomes infinite?	10	3	3
5.	List and explain the characteristics of Markov Chain	10	2	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz	Max Marks	02	04	04	–	02	08	–	–	–	–
	Test	Max Marks	10	20	20	–	10	20	20	–	–	–

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